MATH4341 Lecture Notes

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Contents

1 Lecture 9 - 23/09/26 **2**

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Definition 1.1. A *subbasis* $\mathcal{C}$ for a topology on X is a collection of subsets that cover X, meaning that their union is all of X.

Example 1.2. Given $f : \mathbb{R} \to \mathbb{R}^{\mathbb{N}} = \mathbb{R}^{\omega}$, $f(t) = (t, t, \dots, t, \dots)$ and each $f_i : \mathbb{R} \to \mathbb{R}$, we want

$$f \text{ continuous} \iff f_i \text{ continuous}$$

in other words,

$$f : Y \to \prod_{i \in I}^{\infty} X_i \text{ continuous} \iff \text{each } f_i : Y \to X_i \text{ continous}$$

Claim: $f : \mathbb{R} \to (\mathbb{R}^{\omega}, \text{box topology})$ is **NOT** continuous. Take basis element $B = \prod_{n=1}^{\infty}(-\frac{1}{n}, \frac{1}{n})$ and we claim $f^{-1}(B)$ is not open in $\mathbb{R}$. Note that $x = 0 \in f^{-1}(B)$ since $f(0) = \overrightarrow{0} \in B$. Assume it is open. Then we have an open interval $\prod_{n=1}^{\infty}(-\frac{1}{\epsilon}, \frac{1}{\epsilon}) \in B$ around $x = 0$. But this implies for the box topology $(-\frac{1}{\epsilon}, \frac{1}{\epsilon}) \in (-\frac{1}{n}, \frac{1}{n}) \forall n$ which is not true, so we need a new topology.
Since $f_i = \pi_i \circ f$, we need π_i to be continuous to satisfy the desired condition. So find a topology for which this is true. We want

$$\pi_j : \prod_{i \in I}^{\infty} X_i \to X_j$$

to be continuous, so find a topology to put on the product space.

$$\pi_j^{-1} \text{ is open in } \prod_{i \in I}^{\infty} X_i$$

notice that when projecting onto an open set along an axis for $\mathbb{R}^2$, we have

$$\pi_j^{-1} = U_1 \times X_2$$

for an open set U_1 in X_j. Thus the preimage is a strip where one dimension is unrestricted. The same applies for every other dimension. However, these strips are not a basis because the infinite vertical strips do not satisfy (B2) because the intersection of two strips cannot contain an infinite strip. Thus we create a subbasis so that our finite rectangles are subbasis elements.

$$\pi_j^{-1}(U_j) = \prod_{i \in I}^{\infty} V_i,$$

where

$$V_i = \begin{cases} X_i & \text{if } i \neq j \\ U_j & \text{if } i = j \end{cases}$$

Also note that a basis = finite intersection of a subbasis $= \prod_{i \in I}^{\infty} V_i$ where $V_i = X_i$ for all but finitely many i's. This is the product topology.

$$\textit{subbasis} \to_{\text{finite intersection}} \textit{basis} \to_{\text{union}} \textit{topology}$$

The subbasis allows us to include the missing scalar multiplication operation for vector spaces that we are missing when we begin from just a basis.

Theorem 1.3.

$$f : Y \to \prod_{i \in I}^{\infty} X_i \text{ continuous} \iff \text{each } f_i : Y \to X_i \text{ continous}$$

Proof. ($\Rightarrow$) Assume f is continuous. We want to show $f^{-1}(U)$ is open in Y. It is enough to consider U as a subbasis because subbasis implies basis holds and topology holds. The preimage of the projection is of the form

$$\pi_j^{-1}(U_j) = \prod_{i \in I}^{\infty} U_i$$

where $U_i = X_i$ if $i \neq j$. so then a subbasis element is

$$f^{-1}(\pi_j^{-1}(U_j)) = (\pi_j \circ f)^{-1}(U_j) = f_j^{-1}(U_j) = f_j$$

thus since the subbasis elements are equal to f_j the preimage of each are open in Y thus f_j is continuous.

$f^{-1}(U)$ open in Y for subbasis U implies the basis $B = U_1 \cap \cdots \cap U_n$ which implies $f^{-1}(B) = f^{-1}(U_1) \cap \ldots$ implies open which implies holds true for topology.[1]
($\Leftarrow$) Trivial, in slides. □

[1] Show this at home. Also show that this proof fails for box topology and why we need product topology (to show π_i continuous for forward direction)